Result 1 (identity shape vertex). [Topology foundations, shape maps and diagram pullbacks]
Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let v be an element of V. Then

$$\mathrm{id}(Q).\mathrm{vertices}(v) = v.$$

Result 2 (composite shape vertex). [Topology foundations, shape maps and diagram pullbacks]
Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Let U and J be types. Let S be a directed shape with vertices in U and edges in J. Let g be a map of directed shapes from R to S. Let v be an element of V. Then

$$(g \circ f).\mathrm{vertices}(v) = g.\mathrm{vertices}(f.\mathrm{vertices}(v)).$$

Result 3 (composite shape edge). [Topology foundations, shape maps and diagram pullbacks]
Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Let U and J be types. Let S be a directed shape with vertices in U and edges in J. Let g be a map of directed shapes from R to S. Let e be an element of E. Then

$$(g \circ f).\mathrm{edges}(e) = g.\mathrm{edges}(f.\mathrm{edges}(e)).$$

Result 4 (shape left identity). [Topology foundations, shape maps and diagram pullbacks]
Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Then

$$(f \circ \mathrm{id}(Q)) = f.$$

Result 5 (shape right identity). [Topology foundations, shape maps and diagram pullbacks]
Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Then

$$(\mathrm{id}(R) \circ f) = f.$$

Result 6 (module transport reflexivity). [Topology foundations, shape maps and diagram pullbacks]
Let K and T be types. Let *field* be a field structure on K. Let M be a function from T to Mod_K. Let x be an element of T. Then

$$\mathrm{transport}_M(\mathrm{refl}_x) = \mathrm{id}_{M(x)}\,.$$

Result 7 (pullback vertex module). [Topology foundations, shape maps and diagram pullbacks]
Let K be a type. Let *field* be a field structure on K. Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Let M be a module diagram on R. Let v be an element of V. Then

$$f^*(M).\mathrm{object}(v) = M.\mathrm{object}(f.\mathrm{vertices}(v)).$$

Result 8 (identity pullback). [Topology foundations, shape maps and diagram pullbacks]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let M be a module diagram on Q. Then

$$\mathrm{id}(Q)^*(M) = M.$$

Result 9 (composite pullback). [Topology foundations, composition of diagram pullbacks]
Let K be a type. Let *field* be a field structure on K. Let V, E, W, F, U and J be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let S be a directed shape with vertices in U and edges in J. Let f be a map of directed shapes from Q to R. Let g be a map of directed shapes from R to S. Let M be a module diagram on S. Then

$$(g \circ f)^*(M) = f^*(g^*(M)).$$

LLM-assisted specimens and proofs (gpt-6, 2026-11-06). See PROVENANCE.md. Shape changes do not yet construct the pyramid or its interval-basis restriction.